KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION

SCHOOL OF COMPUTING

COMPUTER SCIENCE AND ENGINEERING

SevaMatch AI

AI-POWERED VOLUNTEER–NGO MATCHING PLATFORM

PROJECT REPORT

Project ID: T054 Team ID: PRJ054

Team Members

Student Name	Registration Number	Individual GitHub Link
Boreddy Jyothi	99230040012	https://github.com/jyothiboreddy1-netizen/GenAI
C. Vijay Durga Prasad	99230040016	https://github.com/vijaydurga-io/GEN-AI
P. Navya	99230040724	https://github.com/pendlinavya2006-beep/Gen-AI
G. Manaswitha	99230040866	https://github.com/manaswithagaddam-cmyk/Gen-AI
Udyala Akhil	99230040789	https://github.com/BARGAV78/GEN-AI

Team GitHub Repository:

https://github.com/jyothiboreddy1-netizen/Sevamatch-ai

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

SCHOOL OF COMPUTING – COMPUTER SCIENCE AND ENGINEERING

KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION

KRISHNANKOIL – 626 126

November 2026

PROJECT TRAINER / GUIDE

Mr.SHIRWIN PRINCE I

Table of Contents

Section	Title
1	Introduction
2	Problem Statement & Objectives
3	System Architecture (AI Matching and RAG Pipeline)
4	Implementation Details & Tech Stack
5	Results and Testing
6	Conclusion & Future Scope

1. Introduction

Volunteering plays an important role in community development by connecting people who are willing to contribute their time and skills with organizations working for social causes. However, volunteers may find it difficult to identify opportunities that match their skills, interests, location and availability. At the same time, non-governmental organizations (NGOs) need an efficient way to publish opportunities and reach suitable volunteers.

To address these challenges, SevaMatch AI – an AI-Powered Volunteer–NGO Matching Platform is developed as a web-based application that helps volunteers discover relevant NGO opportunities and supports NGOs in managing their activities and applicants.

The platform collects volunteer preferences such as skills, interests, location and availability. It maintains an NGO opportunity dataset containing details such as the cause, participation mode, location, time commitment and required skills. The system uses these details to identify relevant opportunities and applies location and availability filters to improve discovery.

SevaMatch AI also integrates Generative AI through the Google Gemini API. A Retrieval-Augmented Generation (RAG)-oriented workflow uses relevant opportunity information as context for AI-assisted guidance. The application is developed using Python and FastAPI for backend services, HTML, CSS and JavaScript for the frontend, and MongoDB for data storage.

The primary objective of SevaMatch AI is to make volunteer discovery and NGO coordination easier, more relevant and more accessible through preference-based matching and AI-assisted interaction.

2. Problem Statement & Objectives

Problem Statement

Volunteers often depend on scattered websites, social media posts and personal networks to find volunteering opportunities. Searching manually can take considerable time, and the available opportunities may not match an individual's skills, interests, location or schedule. As a result, volunteers may miss relevant opportunities or spend time exploring unsuitable listings.

NGOs also face difficulties in reaching volunteers who have the skills and availability required for a particular activity. Managing opportunity information and reviewing applicants manually can make coordination less efficient. A centralized platform is therefore needed to connect volunteers and NGOs through structured opportunity information and relevant matching.

SevaMatch AI addresses this problem by combining volunteer preference collection, NGO opportunity management, skill and interest matching, location and availability filtering, application workflows and AI-assisted guidance in a single web-based application.

Objectives

The main objective of the SevaMatch AI project is to develop an intelligent platform that connects volunteers with suitable NGO opportunities and simplifies opportunity management.

- Volunteer Profiles: To collect volunteer details, skills, interests, location and availability.
- NGO Opportunity Dataset: To maintain structured information about NGO opportunities and their requirements.
- Skill and Interest Matching: To compare volunteer preferences with opportunity requirements.
- Location Filtering: To help volunteers discover opportunities within a suitable geographical radius.
- Availability Filtering: To identify opportunities that align with volunteer time availability.
- Application Management: To allow volunteers to apply and NGOs to review applications.
- AI-Powered Guidance: To integrate the Google Gemini API for contextual assistance using relevant opportunity information.
- User-Friendly Interface: To provide separate workspaces for volunteers and NGO administrators.

3. System Architecture (AI Matching and RAG Pipeline)

SevaMatch AI follows a web application architecture that connects volunteers and NGO administrators through a browser-based interface. The frontend accepts user inputs and communicates with the backend through API requests. The Python FastAPI backend manages user and opportunity workflows, validates requests, performs matching and filtering operations, and communicates with the MongoDB database.

Volunteer profile information, including skills, interests, location and availability, is compared with the requirements of NGO opportunities. The matching process helps identify relevant opportunities, while location and availability filters narrow the results according to the volunteer's preferences.

For AI-assisted guidance, the system follows a Retrieval-Augmented Generation (RAG)-oriented process. Relevant opportunity information is retrieved from the available platform data and supplied as context together with the user's query. The Google Gemini API uses this context to generate a response, which is then returned to the user interface.

The major stages of the architecture are:

- Input: Volunteers and NGO administrators interact with the SevaMatch AI web interface.
- Backend Processing: FastAPI handles user requests, profile information and opportunity operations.
- Data Storage: MongoDB stores application data such as profiles, NGO information, opportunities and applications.
- Matching and Filtering: Volunteer skills and interests are compared with opportunity requirements, with location and availability filters applied.
- Context Retrieval and AI: Relevant opportunity details are provided as context to the Gemini API for AI-assisted guidance.
- Output: Matched opportunities, application information and AI-generated guidance are displayed in the interface.

SevaMatch AI – System Architecture

Volunteer–NGO Matching and AI Guidance Pipeline

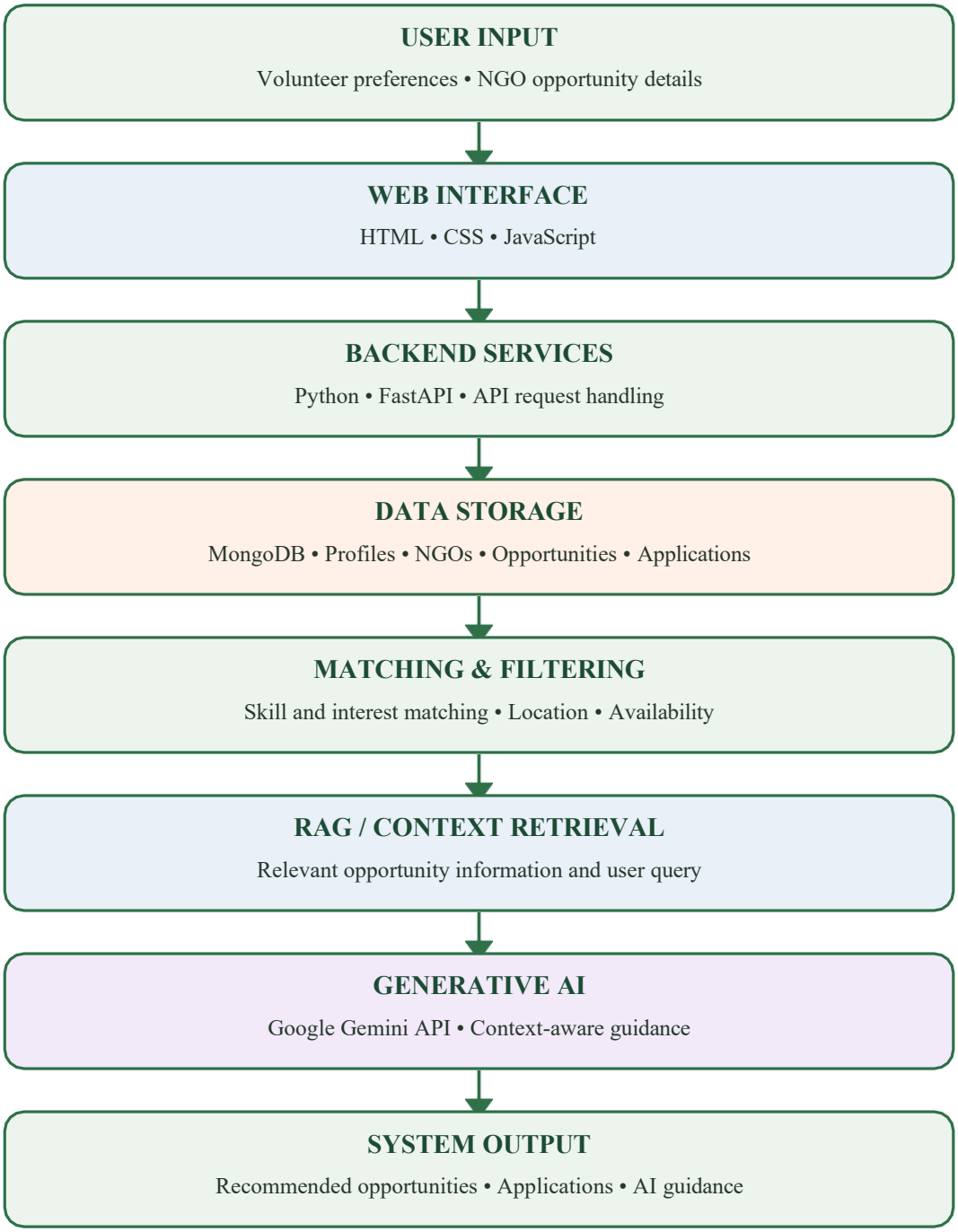


Fig. 3.1: Overall system architecture of SevaMatch AI

The architecture illustrates the flow of information from volunteer and NGO inputs to backend processing, database access, matching, retrieval-grounded AI assistance and final output. The diagram represents the logical system design based on the project's described technologies and implemented screens.

3.1 Data Ingestion and Profile Management Phase

The data ingestion phase begins when volunteers and NGO administrators enter information through the SevaMatch AI interface. Volunteers provide profile details and preferences such as skills, interests, location and availability. These details are used by the application to identify suitable opportunities.

NGO administrators provide organization information and publish volunteering opportunities. The opportunity form captures information such as opportunity title, description, cause, participation mode, location, hours required per week and required skills. These records form the opportunity dataset used by the discovery and matching features.

The backend processes the submitted information and stores the required application records in MongoDB. This structured information supports opportunity browsing, profile management, application handling and AI-assisted retrieval.

3.2 Query, Matching and Generation Phase

The query and matching phase begins when a volunteer explores available opportunities or requests guidance. The system compares the volunteer's skills and interests with the opportunity requirements. Location and availability filters are used to narrow the results and help the volunteer focus on relevant opportunities.

For AI-assisted guidance, the user's query is processed by the backend and relevant opportunity information is retrieved from the available data. The retrieved context is combined with the query and sent to the Google Gemini API. The generated response is returned to the SevaMatch AI interface to help the user understand available opportunities.

NGO administrators can use the NGO workspace to publish opportunities and view submitted applications. The applicant management interface provides actions to review, accept or decline applications. This completes the main workflow between volunteers and NGOs.

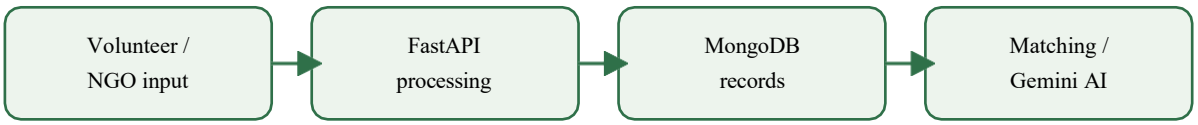


Fig. 3.2: Simplified data and request flow

4. Implementation Details & Tech Stack

SevaMatch AI is implemented using a combination of frontend, backend, database and Generative AI technologies. Each component supports a specific part of the application, from user interaction and API processing to persistent data storage and AI-assisted guidance.

Component	Technology Used	Purpose
Programming Language	Python	Backend application logic and matching-related processing.
Backend Framework	FastAPI	Creating API endpoints and handling frontend requests.
Frontend Technologies	HTML, CSS, JavaScript	Building the user interface, page layout and client-side interactions.
Database	MongoDB	Storing volunteer profiles, NGO details, opportunities and applications.
Generative AI Integration	Google Gemini API	Generating AI-assisted responses using relevant context.
AI Architecture	Retrieval-Augmented Generation (RAG) approach	Retrieving relevant opportunity information and providing it as context to the language model.
Matching	Skill and interest matching	Comparing volunteer preferences with NGO opportunity requirements.
Filtering	Location and availability filters	Narrowing opportunities according to distance and time availability.
API Communication	REST API endpoints	Connecting the frontend with backend application services.
Version Control	GitHub	Managing and maintaining project source code.

The technologies listed above are based on the project details provided by the team. The report does not specify an exact Gemini model version, embedding model, vector database or cloud hosting provider; these implementation details are therefore not assumed.

5. Results and Testing

The SevaMatch AI application was developed as a web-based platform to connect volunteers with NGO opportunities. The application provides separate workspaces for volunteers and NGO administrators. Volunteers can explore opportunities and view recommendations, while NGOs can publish opportunities and manage applicants. The system also includes AI-assisted guidance using the Gemini API.

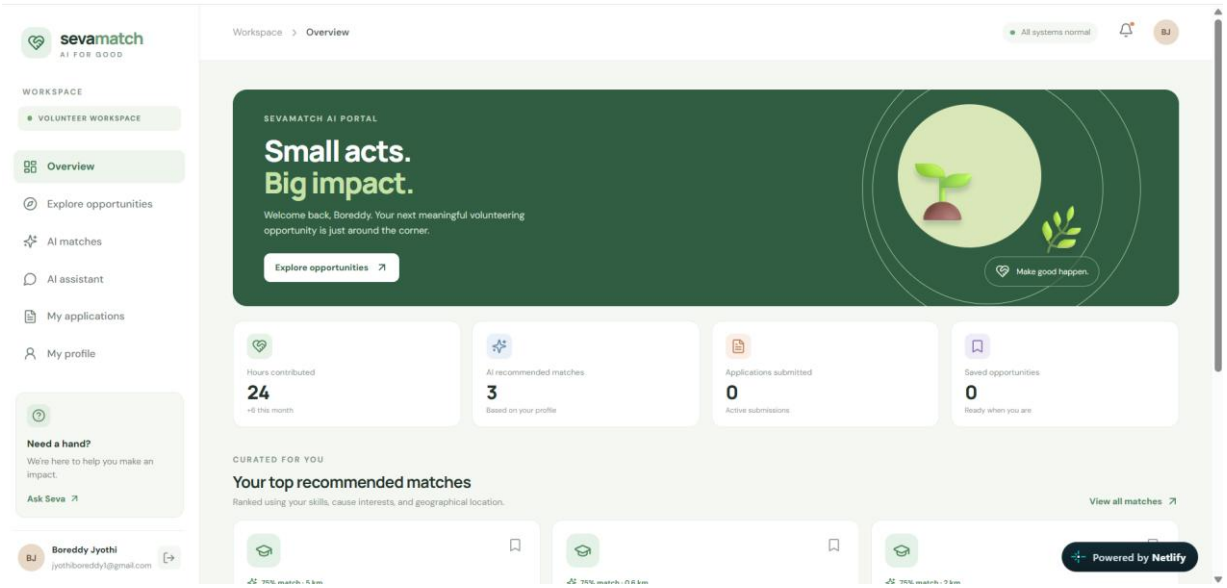


Fig. 5.1

The volunteer dashboard displays the SevaMatch AI home interface, including opportunity recommendations and summary information. It provides an entry point for volunteers to discover relevant volunteering activities.

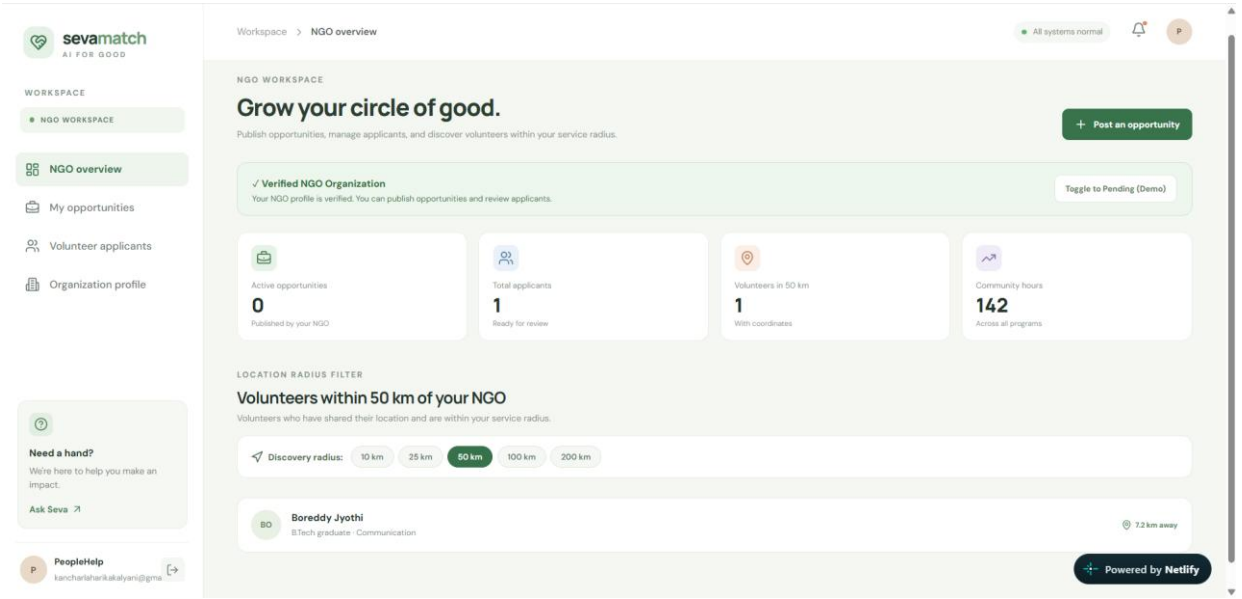


Fig. 5.2

The NGO overview dashboard displays organization-side information and opportunity management options. It includes a volunteer overview and a distance-radius selection that supports location-based discovery.

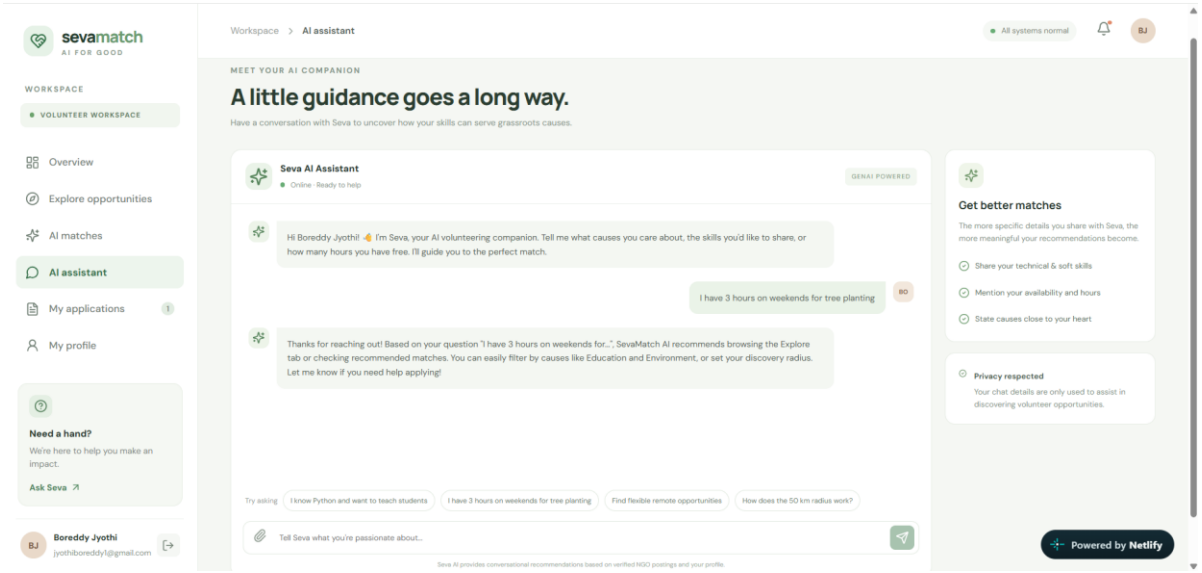


Fig. 5.3

The AI guidance interface provides a conversational space for users to seek assistance. The project integrates the Google Gemini API and uses relevant opportunity information as context for AI-assisted responses.

6. Conclusion & Future Scope

Conclusion

The SevaMatch AI project provides a web-based solution for connecting volunteers with NGOs through preference-based opportunity discovery. It collects volunteer skills, interests, location and availability, maintains NGO opportunity information, and supports matching and filtering to help users identify relevant activities. The platform also provides application management for volunteers and NGOs, along with AI-assisted guidance through the Google Gemini API and a RAG-oriented workflow.

The application is developed using Python, FastAPI, HTML, CSS, JavaScript and MongoDB. The screenshots demonstrate the volunteer dashboard, NGO overview and AI guidance interface. Overall, SevaMatch AI aims to make volunteering opportunity discovery and NGO coordination more organized, accessible and interactive.

Future Scope

- Improved Recommendation: Introduce a formally evaluated ranking score using skill overlap, interest similarity, location and availability.
- Notifications: Add alerts for new opportunities and application status updates.
- Enhanced AI Guidance: Improve contextual retrieval and provide more personalized opportunity explanations.
- Multilingual Support: Support multiple languages to reach a wider volunteer community.
- Analytics Dashboard: Provide NGOs with statistics on applications, volunteer participation and opportunity reach.

Overall: SevaMatch AI can be further developed into a broader volunteer engagement platform that supports NGOs and volunteers through intelligent discovery, effective coordination and responsible AI assistance.